

MICROECONOMICS - OPEN-NOTES EXAM GUIDE

Explained edition for someone who has not read the slides. Built from the supplied IIMA lectures/revision notes, auction notes and the 2018 final. The aim is not just to name a model, but to explain what it means, why the formula works, and how to recognize it inside an unusual story.

1. HOW TO ATTACK A RANDOM-LOOKING QUESTION

Your professor often changes the story while keeping the economic skeleton familiar. Before calculating anything, identify four things: **who chooses, what they choose, when they choose, and what determines payoff**. Once those are clear, the correct model usually becomes obvious.

Clue in story	Underlying model	What to do first
Players act without observing each other	Simultaneous game / Nash	Write or mark each player's best response
One player acts, the next observes	Sequential game / Stackelberg	Solve the later mover first; work backward
Firms choose quantities	Cournot or Stackelberg	Write profit as a function of q ; derive reaction function
Firms choose prices	Bertrand / differentiated price competition	Write demand as a function of own and rival price
Winner pays own bid	First-price / Dutch auction	Trade off probability of winning against profit margin
Winner pays runner-up bid	Second-price / English auction	Truthful bidding is weakly dominant in standard private-value case
Entry fee + per-unit fee	Two-part tariff	Use consumer surplus as the participation constraint
Different observable groups pay different prices	Third-degree price discrimination	Equalize marginal revenues across served groups
Type/quality hidden before contract	Adverse selection / signaling	Ask how types can be separated or screened
Effort/action hidden after contract	Moral hazard / principal-agent	Write incentive-compatibility and participation constraints

Six-step answer protocol: 1) Name the model and timing. 2) Write payoff/profit. 3) State the equilibrium/optimality condition. 4) Solve. 5) Check deviations, participation and boundaries. 6) Translate the number back into the story. A strange story does not require a strange method.

EXAM TRAP: Do not jump from a keyword to a memorized answer. Example: "firms compete in prices" does *not* automatically imply price = marginal cost. The standard Bertrand result needs homogeneous products and the usual identical-cost assumptions.

What the 2018 final tells us about style

The supplied final moves rapidly from monopsony to education signaling, a doctor incentive contract, the 11-20 game, insurance, pollution, transfer pricing, joint products, advertising, two-part tariffs, price discrimination, differentiated-price competition, bundling and the Lerner index. This means recognition and setup matter at least as much as memorizing definitions.

2. GAME THEORY: BEST RESPONSES, NASH, DOMINANCE, MAXIMIN

The basic idea

A strategic situation is one where your payoff depends not only on what you do, but also on what someone else does. A **best response** is the action that gives you the highest payoff for a particular action of the opponent. A **Nash equilibrium** is a profile in which every player is already playing a best response to the others. Therefore nobody wants to deviate alone.

How to solve a payoff matrix

For the row player, inspect each column and mark the row giving the larger first payoff. For the column player, inspect each row and mark the column giving the larger second payoff. Any cell marked for both players is a pure-strategy Nash equilibrium. If no cell is mutual best response, there is no pure Nash equilibrium; do not invent one.

Dominant strategy: an action that is better than the alternatives regardless of what the opponent does. If both players have dominant strategies, their intersection is a dominant-strategy equilibrium. Dominance is stronger than Nash: a Nash action only needs to be best against the equilibrium action of the opponent.

Maximin: for each own action, look at the worst payoff the opponent could leave you with; then choose the action with the best of those worst payoffs. It is a defensive criterion and need not coincide with Nash.

WORKED EXAMPLE: In the Prisoner's Dilemma from the slides, "confess" is dominant for each prisoner, so both confess in equilibrium even though mutual denial gives both a better joint outcome. This is the classic conflict between individual incentives and collective welfare.

Professor example: the 11-20 game

Each player chooses an integer from 11 to 20 and receives that number. If a player chooses exactly one less than the opponent, that player receives an extra 20. The 2018 final gives **no pure-strategy Nash equilibrium**. Why? Whatever pair you propose, at least one player can typically change the chosen number to improve payoff. The lesson is to check unilateral deviations rather than searching for a familiar-looking cell.

When the rule is modified so that choosing the same number gives both players an extra 10, the supplied answer becomes **(11,11)**. Again, the equilibrium changes because the payoff structure changes; never carry over the old equilibrium mechanically.

Sequential games and credible threats

If a later player observes the earlier move, solve by **backward induction**. At the last decision node, choose what the later player would actually prefer. Then the first player anticipates that response. A threat that would be irrational to carry out once the node is reached is not credible and should not affect the solution.

EXAM TRAP: Simultaneous game: intersect best responses. Sequential game: solve the follower first and substitute backward. Treating a sequential game as simultaneous can completely reverse the answer.

3. OLIGOPOLY: COURNOT, STACKELBERG, BERTRAND

Cournot: firms choose quantities simultaneously

Each firm chooses output while treating the rival's output as given. Market price depends on total output $Q = q_1 + q_2$. Firm i maximizes **profit** $= P(Q)q_i - C_i(q_i)$. The first-order condition gives a **reaction function**: the firm's profit-maximizing quantity for every possible rival quantity. The Cournot Nash equilibrium is where the reaction functions intersect.

WORKED EXAMPLE: Slide example: $P = 20 - Q$, $MC_1 = MC_2 = 4$. Firm 1 profit is $[20 - (q_1 + q_2)]q_1 - 4q_1$. FOC gives $R_1(q_2) = 8 - q_2/2$; similarly $R_2(q_1) = 8 - q_1/2$. Solving together gives $q_1^* = q_2^* = 16/3$, total $Q = 32/3$, $P = 28/3$, profit $256/9$ each. Under collusion, the firms act like one monopolist: $q_1 = q_2 = 4$, $P = 12$, profit 32 each.

Linear shortcut: If $P = a - bQ$ and both firms have constant $MC = c$, then $R_i = (a - c)/(2b) - q_j/2$ and the symmetric Cournot output is $q_i = (a - c)/(3b)$. Use this only when the assumptions match.

Stackelberg: quantities chosen sequentially

The follower sees the leader's quantity and then chooses its best response. Therefore the leader can predict the follower's reaction and deliberately choose a quantity that moves the follower along its reaction curve. Procedure: 1) derive follower $R_2(q_1)$; 2) substitute $q_2 = R_2(q_1)$ into leader profit; 3) maximize leader profit; 4) recover follower quantity.

WORKED EXAMPLE: With the same $P = 20 - Q$ and $MC = 4$ example, the follower reaction is $q_2 = 8 - q_1/2$. Substituting this into leader profit gives $q_1^* = 8$. Then $q_2^* = 4$, $P = 8$, leader profit = 32 and follower profit = 16. This is the slide's first-mover advantage.

Bertrand: firms choose prices

For homogeneous goods with identical constant marginal cost, a firm charging above its rival loses the market; a firm can profitably undercut any common price above MC. This pressure continues until the standard equilibrium is $p_1 = p_2 = MC$. The economic logic is **undercutting**, not a quantity reaction function.

With **differentiated products**, consumers may still buy from a higher-priced firm because they prefer its brand. Then each firm has demand $Q_i(p_i, p_j)$, and you maximize $(p_i - MC_i)Q_i$. Positive markups can survive. This is why the 2018 Danti-Kanti toothpaste problem is not the plain homogeneous-Bertrand model.

Model	Strategic variable	Timing	Core solution
Cournot	Quantity	Simultaneous	Derive quantity BRs; solve together
Stackelberg	Quantity	Sequential	Follower BR $\rightarrow$ leader optimization
Bertrand, homogeneous	Price	Simultaneous	Undercutting $\rightarrow p = MC$ under standard assumptions
Differentiated-price competition	Price	Usually simultaneous	Demand depends on own and rival price; derive price BRs

EXAM TRAP: "First mover advantage" is not universal. In some price-competition settings moving first can be disadvantageous. Always solve the follower's actual response rather than assuming leadership is valuable.

4. AUCTIONS: WHAT EACH FORMAT CHANGES

Format	How it works	Strategic implication
First-price sealed bid	Highest bid wins; winner pays own bid	Bid below value: higher bid raises win chance but reduces margin
Second-price / Vickrey	Highest bid wins; pays second-highest bid	Truthful bidding is weakly dominant in standard private values
English ascending	Price rises until all but one quit	Strategically like second-price: stay until price reaches value
Dutch descending	Price falls until someone accepts	Strategically like first-price: acceptance threshold acts like own bid

Why truth-telling works in a second-price auction

Suppose your value is v . Your own bid determines whether you win, but if you win, the price is determined by the highest rival bid. Bidding above v can only matter when it makes you win at a price above your value, creating a loss. Bidding below v can only matter when it makes you lose an item you could have bought profitably. Therefore bidding v is weakly dominant.

Under truthful bidding, the highest-value bidder wins and pays the second-highest value, so the allocation is efficient. But **weakly dominant does not mean unique Nash equilibrium**. The notes deliberately show other complete-information equilibria.

First-price auction: why you shade

If you bid your full value in a first-price auction, winning gives zero surplus. So you normally bid below value. A higher bid makes winning more likely but leaves less profit conditional on winning. The equilibrium balances these two forces.

Standard formula: with n risk-neutral bidders, independent values uniformly distributed on $[0,1]$, and a symmetric increasing equilibrium, $b(v) = ((n-1)/n)v$. With two bidders, bid $v/2$. More bidders means less shading because competition is stronger.

Derivation idea: pretend you bid as though your value were z . If everyone else uses an increasing strategy, you win only if all $n-1$ rival values are below z , so $\Pr(\text{win}) = z^{n-1}$. Maximize $z^{n-1}[v - b(z)]$ and impose $z = v$ in equilibrium.

Known values / discrete bids: do not use the uniform formula

WORKED EXAMPLE: Auction notes: values 2,4,6,8,10 are common knowledge, bids are integers, ties favor the higher-value bidder. In first price, the value-10 bidder can bid 8, beat/tie the value-8 bidder under the rule, and earn 2. In second price, the notes exhibit a Nash equilibrium where the value-10 bidder bids 10 and others bid 1, so the winner pays 1 and earns 9. The point is to test deviations directly.

Revenue equivalence and winner's curse

Under the standard assumptions of risk neutrality, independent private values and symmetry, the four auction formats have the same expected seller revenue. This result can fail with risk aversion or correlated/common values.

Winner's curse: when the object has an uncertain common component, winning is information. The revision example has startup value v uniformly distributed from 10 to 30 and an offer $B=20$. Acceptance means $v \leq 20$, so the conditional expected value is 15, not the unconditional 20. Always condition on the event that made you win or get accepted.

EXAM TRAP: Ask first: private value or common/value uncertainty? Known values or random values? Own-price or second-price payment? Those details determine the strategy.

5. PRICING WITH MARKET POWER: FROM MONOPOLY TO DISCRIMINATION

Uniform monopoly and the Lerner condition

A monopolist chooses quantity where $MR=MC$ and then reads the selling price from demand. Because the demand curve slopes down, marginal revenue lies below price. The Lerner index links markup to elasticity: $(P-MC)/P = 1/|\epsilon|$. Less elastic demand supports a larger markup.

WORKED EXAMPLE: 2018 final: $P=10$ and $MC=8$. Markup ratio $= (10-8)/10 = 0.2 = 1/|\epsilon|$, so $|\epsilon|=5$.

First-degree price discrimination

The seller charges each buyer (or each unit) its reservation value. Conceptually, the demand curve becomes the seller's marginal revenue curve because the seller does not need to lower the price on earlier units to sell one more unit. Output expands until marginal willingness to pay equals MC . Under perfect first-degree discrimination, the firm captures the consumer surplus.

WORKED EXAMPLE: Slides: $Q=10-P$ and $C(Q)=Q^2$. Uniform monopoly: $MR=10-2Q$, $MC=2Q \rightarrow Q=2.5$, $P=7.5$, profit=12.5. Perfect first-degree PD: use demand/ MB $10-Q=MC=2Q \rightarrow Q=10/3$; profit rises to 16.66.

Second-degree price discrimination

The firm cannot directly identify types, so it offers a menu based on quantity or version and lets customers self-select. Examples in the material include slab/block tariffs and versions. The important idea is that price depends on the package chosen rather than an observed demographic label.

Third-degree price discrimination

The firm can separate observable groups (students, tourists, seniors, locations, etc.) and prevent resale. It chooses sales in each segment so the marginal revenue from the last unit is equal across segments and equal to common MC : $MR_1=MR_2=\dots=MC$. A segment with less elastic demand receives a higher markup.

WORKED EXAMPLE: 2018 music-lesson question: general sessions cost Rs 600, separate PGP sessions Rs 900. Because the seller is separating an identifiable group and charging it differently, this is third-degree price discrimination. The follow-up asks for the implied elasticity, using the markup-elasticity relationship.

Intertemporal vs peak-load pricing

Intertemporal: consumers differ in patience. Firms charge a high early price to inelastic early adopters and a lower later price to the elastic mass market.

Peak-load: demand itself differs across peak and off-peak periods, so the efficient/profit-maximizing price can differ by time because the relevant demand and capacity conditions differ.

Question clue	Classification
Exact reservation value / haggling buyer by buyer	First degree
Slab, quantity block, version menu	Second degree
Student / tourist / senior / location group	Third degree
Early launch vs later mass market	Intertemporal
Peak vs off-peak demand	Peak-load

6. TWO-PART TARIFFS, BUNDLING, TYING, ADVERTISING, JOINT PRODUCTS

Two-part tariff: entry fee T plus per-unit price P

With one identical customer type, set $P=MC$. That maximizes the customer's consumer surplus, and then set the entry fee equal to that surplus. The customer is just willing to participate, while the firm captures the surplus through T. Thus the fixed fee does the extraction and the per-unit price preserves efficient usage.

With heterogeneous customers, one common entry fee cannot extract everyone's surplus. For any chosen P and set of customers you want to keep, the highest feasible T is the **smallest consumer surplus among those customers**. You must compare serving everyone at a low fee with excluding low-demand types and charging a higher fee.

WORKED EXAMPLE: 2018/revision: $Q_1=100-P$ and $Q_2=160-P$, one million of each, $MC=0$. Serving only type 2 with $P=0$ gives $CS=0.5(160)(160)=12,800$, so $T=12,800$ and profit 12,800 million. Serving both forces $T \leq 5,000$, giving only 10,000 million. Hence exclusion is better in this example.

Bundling

Bundling sells two products together. It works especially well when reservation values are **negatively correlated**: a buyer who values A highly tends to value B less, and vice versa. Then total bundle values are more similar across buyers, allowing a common bundle price to extract more revenue.

For each consumer, compute surplus from every available option: nothing, A alone, B alone, and the bundle. The consumer chooses the highest nonnegative surplus. Never decide based only on whether the consumer can "afford" the bundle.

WORKED EXAMPLE: 2018 dinner: entree price 1500, dessert 500, bundle 1800; values are 1900 and 100. Surplus from entree alone=400; dessert=-400; bundle=200. The diner buys the entree alone, despite valuing the bundle above its price.

Tying

Tying means buying one product requires buying another, such as printer + cartridges or a franchisee sourcing inputs from the parent. Economically it can act like a metered two-part tariff: heavy users buy more of the tied product and therefore pay more overall.

Advertising

The notes use $A/(P \cdot Q) = E_A/E_P$. The firm spends a larger share of sales on advertising when advertising is more effective (higher advertising elasticity) and a smaller share when demand is very price elastic. The 2018 trap reverses the first relationship: the advertising-to-sales ratio does *not* rise with price elasticity.

Joint products

Some production processes generate goods in fixed proportion, such as milk producing butter and buttermilk. There is one common production quantity, so add the products' marginal revenues vertically and set the total equal to MC: $MR_1 + MR_2 = MC$. If one product's MR is negative at that quantity, treat its contribution as zero: it can be better to discard excess than sell where extra sales reduce revenue.

WORKED EXAMPLE: 2018: butter demand $Q=100-P$ and buttermilk $q=80-p$, $MC=20$. $MR_b=100-2Q$ and $MR_m=80-2Q$, so total $MR=180-4Q$. Set $180-4Q=20 \rightarrow Q=40$. Butter price= $100-40=60$.

7. TRANSFER PRICING: THINK “WHERE IS THE NEXT ENGINE WORTH MOST?”

Setup and key object

A vertically integrated firm has an upstream division producing an intermediate input (e.g., engines) and a downstream division converting it into a final product (cars). Define $NMR = MR_{\text{final}} - MC_{\text{assembly}}$. NMR is the marginal value of one additional engine when used internally to make another car.

Case	Optimality condition	Meaning
No outside engine market	$NMR = MC_{\text{up}}$	Make engines until the last engine costs what it adds downstream
Competitive outside market	$NMR = MC_{\text{up}} = P_E$	Equalize value of internal use, in-house production and market trade
Monopoly outside engine market	$MC_{\text{up}} = NMR = MR_E < P_E$	Allocate engines between internal cars and monopoly external sales by marginal revenue

Case 1: no outside market

Every engine made must be used internally. Compute NMR from the final-product demand and downstream processing cost, compute upstream MC, and set them equal. Because one engine becomes one car in the standard examples, the quantities coincide.

WORKED EXAMPLE: Slide example: car demand $Q=520-P$, assembly cost Q^2 , engine cost $2Q_E^2$ and $Q=Q_E$. The supplied FOC gives $Q_E=65$, so 65 cars are made and $P=520-65=455$.

Case 2: competitive outside market

The firm can buy or sell engines at fixed price P_E . Two separate decisions appear. First, how many engines should the upstream division make? Produce until $MC_{\text{up}}=P_E$. Second, how many cars should the downstream division make? Use engines until $NMR=P_E$. The difference between desired use and in-house production determines buying or selling.

WORKED EXAMPLE: Revision example: at $P_E=2000$, in-house production is 1000 but optimal final output needs 1500 engines, so buy 500. At $P_E=4000$, make 2000 but use only 500 internally, so sell 1500. This is why “engines made” and “cars sold” must not be confused.

Case 3: firm is a monopoly seller in the outside engine market

Now selling one more engine externally requires moving down an outside demand curve, so the relevant outside return is MR_E , not the outside price itself. Allocate output until $MC_{\text{up}}=NMR=MR_E$. Because monopoly marginal revenue lies below price, P_E can exceed the internal transfer price.

Unifying result from the revision notes: at an interior optimum, the internal transfer price equals upstream marginal cost in all three cases. The outside market price equals that marginal cost only when the outside market is competitive.

EXAM TRAP: Do not automatically set transfer price equal to outside price. In the monopoly outside-market case, the outside price is above the common marginal value; the internal transfer price is tied to upstream MC.

8. ASYMMETRIC INFORMATION: ADVERSE SELECTION, SIGNALING, MORAL HAZARD

The sorting question

Ask: **Is the hidden thing a fixed type before the deal, or a chosen action after the deal?** Hidden type/quality before contracting is adverse selection. Hidden effort/action after contracting is moral hazard. This one distinction classifies many apparently unrelated questions.

Adverse selection and lemons

When sellers know quality but buyers do not, buyers may only be willing to pay an average value. High-quality sellers may then withdraw, lowering average quality further and causing the market to unravel. The slide example has peaches worth 4L and lemons 2L with 80% peaches: initial expected value is 3.6L, but at that price peaches leave, so buyers eventually offer only 2L.

Signaling

A signal separates types only if it is easier/cheaper for the high type to produce. Suppose obtaining signal level y costs $c_H y$ for high type and $c_L y$ for low type, with $c_H > c_L$; to keep the high type willing, $c_H y^* \leq \Delta$. Thus $\Delta/c_L < y^* \leq \Delta/c_H$.

With three types, as in the 2018 education question, compare the salary gain from moving to the next category with the incremental education costs for each type. Choose thresholds that make the intended type willing to cross but the lower type unwilling to imitate.

Moral hazard / principal-agent

The principal cannot observe effort directly but can observe an outcome such as revenue. A wage scheme should make the desired effort privately optimal for the agent. Write **IC** constraints: utility from desired effort must be at least utility from every alternative. Also check **IR**: the agent must prefer accepting the contract to the outside option.

WORKED EXAMPLE: 2018 clinic: effort costs $H/M/L=30/20/0$ and success probabilities $0.8/0.5/0.2$ for revenue 100. Let $\Delta w = w(100) - w(0)$. H vs M requires $0.3 \Delta w \geq 10$, so $\Delta w \geq 33.33$. H vs L requires $0.6 \Delta w \geq 30$, so $\Delta w \geq 50$. The tight constraint is H vs L , hence the required wage gap is 50. The supplied maximum expected owner profit is 40.

EXAM TRAP: The "next-best" alternative is the one that produces the tightest IC constraint, not necessarily the middle effort level. Compare desired effort against every alternative.

Hidden problem	Timing	Typical fix
Adverse selection	Before deal; hidden type/quality	Signaling, screening, verified records, warranties, pooling
Moral hazard	After deal; hidden action/effort	Monitoring, incentive pay, outcome-contingent contract

9. RISK, INSURANCE, EXTERNALITIES, PUBLIC GOODS

Risk: expected value is not expected utility

Expected value is $E(X) = \sum p_i X_i$. A risk-averse person evaluates uncertain wealth using expected utility $EU = \sum p_i u(X_i)$, where u is concave. Concavity means an extra rupee is worth less when already wealthy, so a sure amount can be preferred to a gamble with the same expected money value.

Certainty equivalent (CE) is the sure wealth giving the same utility as the gamble: $u(CE) = EU$. **Risk premium** $= E(X) - CE$. For a risk-averse person it is positive.

Insurance: fair premium vs maximum willingness to pay

For full insurance against loss L with probability p , the actuarially fair premium is pL : expected premium equals expected payout. A risk-averse consumer may pay more. The maximum premium solves utility of certain insured wealth equal to expected utility without insurance. If initial wealth is W , $u(W - \text{premium_max}) = EU(\text{no insurance})$.

WORKED EXAMPLE: 2018 jewelry store: $W=64L$; theft probability .2; theft leaves wealth $16L$; $u = \sqrt{W}$. $EU = .8(8) + .2(4) = 7.2$. Therefore $CE = 7.2^2 = 51.84$. Maximum premium $= 64 - 51.84 = 12.16L$.

Externalities and corrective policy

An externality is a cost or benefit imposed on a third party but not reflected in the private market price. Negative externality $\rightarrow$ private decision ignores harm $\rightarrow$ too much activity. Positive externality $\rightarrow$ private decision ignores benefit $\rightarrow$ too little activity. The efficient quantity satisfies **MSB=MSC**, where $MSC = \text{private } MC + MEC$ and $MSB = \text{private } MB + MEB$.

A Pigouvian tax for a negative externality is set equal to the marginal external cost at the efficient quantity. For emissions, the efficient target is not zero pollution: set marginal external cost of emissions equal to marginal cost of abatement.

WORKED EXAMPLE: 2018 petrol: supply $Q=P$, demand $Q=300-P$, external cost $C=Q$. The supplied solution gives efficient $Q=100$ and a corrective tax of 100 per unit.

Tradeable permits and Coase

With tradeable permits, firms with low abatement costs reduce pollution and sell permits; high-cost abaters buy permits. Trading continues until marginal abatement costs are equalized, minimizing total cost for a given emissions cap. Under the Coase theorem, clearly defined property rights plus costless bargaining can produce the efficient quantity regardless of who initially owns the right; the initial right affects who pays whom.

Public goods

A public good is non-rival and non-excludable. Everyone consumes the same quantity, so individual marginal benefits are added **vertically**:

$MSB(Q) = MB_A(Q) + MB_B(Q) + \dots$. Set $MSB = MC$. Non-excludability creates free-riding because people hope others will pay while they still enjoy the good.

10. FACTOR MARKETS + FINAL FORMULA / TRAP BANK

Labor demand and marginal revenue product

Hiring one more worker creates MPL extra units of output. Each extra unit contributes marginal revenue MR, so **MRPL=MPL x MR**. If the output market is perfectly competitive, $MR=P$ and $MRPL=MPL \times P$. A competitive input buyer faces a flat wage, so it hires until $MRPL=w$.

Monopsony

A monopsonist has buyer power in the input market and faces an upward-sloping labor supply curve. To hire an additional worker it may need to raise the wage paid to all workers, so **marginal expenditure ME exceeds the wage/average expenditure AE**. Solve employment from $ME=MRPL$, then read the wage from the labor supply curve. If $w=a+bL$, total labor expenditure $= (a+bL)L$ and $ME=a+2bL$.

This is why a monopsonist typically hires fewer workers and pays a wage below the MRPL of the last worker. The height of the ME-MRPL intersection is marginal expenditure, *not* the wage.

Industry labor-demand feedback

For one firm, MRPL is its labor demand. For an industry, simply horizontally adding firms' initial MRPL curves misses a feedback: when all firms hire more, industry output rises, output price falls, and each firm's MRPL shifts. The revision notes flag this as a recurring conceptual trap.

Backward-bending labor supply

A higher wage has a substitution effect (leisure becomes more expensive $\rightarrow$ work more) and an income effect (worker is richer $\rightarrow$ if leisure is normal, consume more leisure and work less). At high wages the income effect can dominate, making labor supply bend backward.

Formula bank

Topic	Formula / condition	Interpretation
Monopoly	$MR=MC$; price from demand	Choose quantity first, then read price
Lerner	$(P-MC)/P=1/ \epsilon $	Less elastic demand $\rightarrow$ larger markup
Cournot	Own-profit FOC $\rightarrow$ reaction function	Solve reaction functions simultaneously
Stackelberg	Follower BR $\rightarrow$ leader profit	Backward induction
FPSB iid uniform	$b(v)=((n-1)/n)v$	Bid shading under standard assumptions
SPSB	Bid value weakly dominant	Own bid affects winning, not price paid
Two-part, one type	$P=MC$; $T=CS$	Efficient usage + surplus extraction
3rd-degree PD	$MR_1=MR_2=\dots=MC$	Allocate sales across groups
Advertising	$A/(PQ)=E_A/E_P$	Ad effectiveness vs price sensitivity
Joint products	Sum positive $MR_i=MC$	Common production quantity
Transfer, competitive outside	$NMR=MC_{up}=P_E$	Trade the gap between making and using
Signaling	$\Delta c_L < y^* \leq \Delta c_H$	Separate high from low type
Risk	$u(CE)=EU$; $RP=E(X)-CE$	Convert risky prospect to sure equivalent
Insurance	fair $=pL$; max $=W-CE$	Fair price vs max WTP
Externality	$MSB=MSC$; tax $=MEC$ at Q^*	Internalize spillover
Public good	Sum $MB_i=MC$	Vertical sum of benefits
Factor demand	$MRPL=MPL \times MR$	Value of one more worker
Monopsony	$ME=MRPL$; wage from supply	Do not read wage from ME curve

Last-minute professor-style traps

1) Model from structure, not vocabulary. 2) Check unilateral deviations in unusual games and complete-information auctions. 3) Truthful SPSB is weakly dominant but need not be the unique Nash equilibrium. 4) Do not use the first-price uniform shading formula when values are known/discrete. 5) With heterogeneous two-part-tariff customers, compare participation/exclusion. 6) In bundling, calculate surplus from every option. 7) For joint products, cap a negative MR contribution at zero. 8) Internal transfer price equals upstream MC at the optimum; an outside monopoly price can be higher. 9) Moral-hazard IC must beat every alternative effort. 10) Fair insurance premium is not the same as maximum willingness to pay. 11) In monopsony, solve L from $ME=MRPL$ and wage from supply.

If stuck: write the decision-maker's payoff explicitly. Most of these topics reduce to "marginal benefit = marginal cost" or "no profitable unilateral deviation," with the story determining what counts as benefit, cost, and information.